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A unified framework for two-grid methods for a class of nonlinear problems

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Abstract
In this article, we present a unified error analysis of two-grid methods for a class of nonlinear problems. We first study the two-grid method of Xu by recasting the methodology in the abstract framework of Brezzi, Rappaz, and Raviart (BRR) for approximation of branches of nonsingular solutions and derive a priori error estimates. Our convergence results indicate that the correct scaling between fine and coarse meshes is given by $h = \mathcal{O}(H^2)$ for all the nonlinear problems which can be written in and applied to the BRR framework. Next, a correction step can be added to the two-grid algorithm, which allows the choice $h = \mathcal{O}(H^3)$. On the other hand, the particular BRR framework with duality pairing, if it is applied to a semilinear problem, allows a higher order relation $h = \mathcal{O}(H^4)$. Furthermore, even the choice $h = \mathcal{O}(H^5)$ is possible with the correction step either on fine mesh or coarse mesh. In addition, elliptic problems with gradient nonlinearities and the Naiver–Stokes equations are considered to illustrate our unified theory. Finally, numerical experiments are conducted to confirm our theoretical findings. Numerical results indicate that the correction step used as a simple postprocessing enhances the solution accuracy, particularly for problems with layers.

Keywords Two-grid finite element method · Error estimates · Nonlinear problems

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1 Introduction

A large number of physical phenomena are governed by partial differential equations. Most of the models have nonlinearities and efficient numerical algorithms are desired to resolve nonlinearities.

Since the pioneering work of Xu [22,23], two-grid methods have been received a lot of attention and successfully applied to approximate various linear and nonlinear problems (see, for example, [1,2,7–10,12,14–16,20,21,24], just to mention a few). The two-grid scheme is based on passing information between finite element equations defined on two grids of different mesh sizes. In the first step, a nonlinear problem itself is solved in a coarse space, i.e., finite dimensional space with coarse grids. In the second step, the nonlinear problem is linearized locally at the solution obtained in the coarse space. Then, the linearized problem is solved in a fine space. Note that a correction step can be added to improve efficiency of the algorithm.

For the presentation, we use the abstract framework of Brezzi, Rappaz, and Raviart (BRR) [4] and its particular framework with duality pairing proposed by Caloz and Rappaz [5]. The framework was designed for approximation of branches of nonsingular solutions for a class of nonlinear problem. The BRR theory applied to the Navier–Stokes equations can be found in Girault and Raviart [11]. Another application can be found in [13], where mixed approximations of scalar elliptic problems with gradient nonlinearities are considered.

The main contribution in this work is that we recast the two-grid algorithm of Xu in the unified setting using the BRR framework and derive a priori error estimates along with unique solvability of the discrete system. Our convergence results indicate that the correct scaling between fine and coarse meshes is given by $h = \mathcal{O}(H^2)$ for all the nonlinear problems which can be written in and applied to the BRR framework. Furthermore, a correction step on the fine mesh can be added to improve the relation between meshes so that $h = \mathcal{O}(H^3)$. We note that existing results on two-grid methods [8,16] often show $h = \mathcal{O}(H^\alpha)$ for some α with $1 < \alpha \leq 2$. Fine scale convergence is retained under the given relations between meshes. On the other hand, the BRR framework with duality pairing allows higher order relations if it is applied to a particular example. For a semilinear problem, a higher order relation $h = \mathcal{O}(H^4)$ with respect to H^1-norm is obtained by exploiting the smoothness conditions of nonlinear problems. Also, the scaling $h = \mathcal{O}(H^5)$ can be achieved with an extra correction either on fine mesh or coarse mesh. In practice, the correction step can be exploited as a simple postprocessing, which enhances the solution accuracy, particularly for problems with layers.

The remainder of this paper is organized as follows. In the next section, we introduce approximation of branches of nonsingular solutions based on the BRR theory. Then, we give a unified presentation of two-grid methods for nonlinear problems under the BRR framework. In Sect. 3, our unified two-grid theory is applied to nonlinear problems such as a semilinear second-order elliptic problem, the elliptic problem with gradient nonlinearities, and the Navier–Stokes equations. The two-grid method based on the

particular BRR framework with duality pairing is presented in Sect. 4 and applications to specific examples are given in Sect. 5. Numerical results are given to support the theory established in the paper in Sect. 6. Some concluding remarks are given in the last section.

2 Two-grid methods for the BRR framework

In this section we start with the abstract approximation theory of Brezzi, Rappaz, and Raviart for nonlinear problems developed in [5] and [11]. Then, we present unified two-grid algorithms to solve resulting nonlinear algebraic equations.

Let X and Y be real Banach spaces with the norms $\|\cdot\|_X$ and $\|\cdot\|_Y$, respectively. Let $\Lambda \subset \mathbb{R}$ be a compact interval and denote $\mathscr{L}(X;Y)$ by the set of all linear and continuous operators from X into Y. The spaces of multilinear mappings are defined by $\mathscr{L}_l(X;Y) = \mathscr{L}(X;\mathscr{L}_{l-1}(X;Y)), \quad l = 2, 3, \ldots$.

First, we consider the following nonlinear problem: find $(\nu, \phi) \in \Lambda \times X$ such that

$$F(\nu, \phi) := \phi + SG(\nu, \phi) = 0, \tag{1}$$

where $S \in \mathscr{L}(Y;X)$ is a linear operator independent of ν and $G : \Lambda \times X \to Y$ is a $\mathcal{C}^1$ mapping. The set $\{(\nu, \phi(\nu)) : \nu \in \Lambda\}$ is called a branch of solutions of (1) if $F(\nu, \phi(\nu)) = 0$ and the map $\nu \to \phi(\nu)$ is continuous from Λ into X. If the Fréchet derivative $D_\phi F(\nu, \phi(\nu))$ of F with respect to ϕ is an isomorphism of X for all $\nu \in \Lambda$, then the branch of solutions $\{(\nu, \phi(\nu)) : \nu \in \Lambda\}$ is called nonsingular.

To approximate problem (1) we introduce an operator $S_h \in \mathscr{L}(Y;X)$ intended to approximate the linear operator S, where $h > 0$ is a real parameter which will tend to zero. The approximate problem of nonlinear problem (1) is to find $\phi^h \in X$ such that

$$F_h(\nu, \phi^h) := \phi^h + S_h G(\nu, \phi^h) = 0. \tag{2}$$

We assume that the following hypotheses.

(H1) $D_\phi G$ is Lipschitz continuous in the sense that there exists a function $\xi \to L(\xi) : \mathbb{R}_+ \to \mathbb{R}_+$, locally bounded, such that for all $\psi \in \overline{B(\phi, \xi)}$, the closed ball with center ϕ and radius ξ,

$$\|D_\phi G(\nu, \phi) - D_\phi G(\nu, \psi)\|_{X;Y} \le L(\xi)\|\phi - \psi\|_X;$$

(H2) There exists another Banach space $Z \hookrightarrow Y$, continuously imbedding, such that $\forall \nu \in \Lambda, \forall \phi \in X$

$$D_\phi G(\nu, \phi) \in \mathscr{L}(X;Z).$$

(H3) For all $y \in Y$

$$\lim_{h\to 0} \|(S - S_h)y\|_X = 0.$$

(H4)

$$\lim_{h\to 0} \|S - S_h\|_{Z;X} = 0.$$

Note that (H4) is a consequence of (H3) when the imbedding of Z into Y is compact. Under the hypotheses, a unique branch of nonsingular solutions of problem (2) and its convergence property are stated below (see e.g. [4] and [5]).

Theorem 2.1 *Assume that (H1)-(H4) hold and that $\{(\nu, \phi(\nu)) : \nu \in \Lambda\}$ is a branch of nonsingular solutions of (1). Then, there exists sufficiently small $\bar{h} > 0$ such that for each $h < \bar{h}$, there are a neighborhood $\mathcal{O}$ of origin in X and a unique $\mathcal{C}^1$-function $\nu \in \Lambda \to \phi^h(\nu) \in X$ such that $\{(\nu, \phi^h(\nu)) : \nu \in \Lambda\}$ is a branch of nonsingular solutions of (2) and that $\phi(\nu) - \phi^h(\nu) \in \mathcal{O}$ for all $\nu \in \Lambda$. Moreover, there exists a positive constant C, independent of h and ν, such that*

$$\|\phi(\nu) - \phi^h(\nu)\|_X \le C\|(S - S_h)G(\nu, \phi(\nu))\|_X \quad \forall \nu \in \Lambda.$$

Now, we introduce numerical algorithms to practically solve nonlinear problem (2). Assume that $\{(\nu, \phi(\nu)) : \nu \in \Lambda\}$ is a branch of nonsingular solutions of (1). For each value of the parameter $h > 0$, we let X_h be a finite-dimensional subspace of X.

Algorithm 1: Xu's two-grid algorithm
Step 1: Solve nonlinear system on coarse mesh
Find $\phi^H(\nu) \in X_H$ such that $F_H(\nu, \phi^H(\nu)) = 0$.
Step 2: Update on fine mesh with one Newton iteration
Find $\phi_h(\nu) \in X_h$ such that $D_\phi F_h(\nu, \phi^H(\nu))(\phi_h(\nu) - \phi^H(\nu)) = -F_h(\nu, \phi^H(\nu))$.

We will take H as mesh size of coarse grid and h as mesh size of fine grid in applications. We assume that the finite element spaces X_H and X_h are nested i.e., $X_H \subset X_h \subset X$. The invertibility of $D_\phi F_h(\nu, \phi^H(\nu))$ in **Step 2** of **Algorithm 1** comes from the assumption that $D_\phi F(\nu, \phi(\nu))$ is an isomorphism.

Lemma 2.2 *Assume that (H1)-(H4) hold and that $\{(\nu, \phi(\nu)) : \nu \in \Lambda\}$ and $\{(\nu, \phi^H(\nu)) : \nu \in \Lambda\}$ are branch of nonsingular solutions of (1) and (2) respectively, then there are $h^* \le \bar{h}$ and $h_* \le \bar{h}$, where $\bar{h}$ is the parameter of Theorem 2.1, such that for all $H < h^*$ and for all $h < h_*$, $D_\phi F(\nu, \phi^H(\nu))$ and $D_\phi F_h(\nu, \phi^H(\nu))$ are isomorphisms of X with*

$$\|D_\phi F(\nu, \phi^H(\nu))^{-1}\|_{X;X} \le 2\|D_\phi F(\nu, \phi(\nu))^{-1}\|_{X;X}, \tag{3}$$

$$\|D_\phi F_h(\nu, \phi^H(\nu))^{-1}\|_{X;X} \le 2\|D_\phi F(\nu, \phi^H(\nu))^{-1}\|_{X;X} \le 4\|D_\phi F(\nu, \phi(\nu))^{-1}\|_{X;X}. \tag{4}$$

Proof For convenience, we omit the argument ν of the functions $\phi(\nu)$ and $\phi^H(\nu)$. Note that

$$D_\phi F(\nu, \phi) - D_\phi F(\nu, \phi^H) = S(D_\phi G(\nu, \phi) - D_\phi G(\nu, \phi^H)).$$

So,

$$\|D_\phi F(\nu,\phi)^{-1}(D_\phi F(\nu,\phi) - D_\phi F(\nu,\phi^H))\|_{X;X} \\ \leq \|D_\phi F(\nu,\phi)^{-1}\|_{X;X}\|S\|_{Y;X}\|D_\phi G(\nu,\phi) - D_\phi G(\nu,\phi^H)\|_{X;Y}.$$

From the hypothesis (H1) and Theorem 2.1, we have

$$\|D_\phi G(\nu,\phi) - D_\phi G(\nu,\phi^H)\|_{X;Y} \leq C\|(S - S_H)G(\nu,\phi)\|_X,$$

which approaches 0 as $H \to 0$ by (H3). Hence, there is $h^*(\leq \bar{h})$ such that for all $H < h^*$,

$$\|D_\phi F(\nu,\phi)^{-1}(D_\phi F(\nu,\phi) - D_\phi F(\nu,\phi^H))\|_{X;X} \leq 1/2.$$

As a consequence

$$D_\phi F(\nu,\phi^H) = D_\phi F(\nu,\phi)\big(I - D_\phi F(\nu,\phi)^{-1}[D_\phi F(\nu,\phi) - D_\phi F(\nu,\phi^H)]\big)$$

is an isomorphism of X and

$$\|D_\phi F(\nu,\phi^H)^{-1}\|_{X;X} \\ \leq \Big\|\big(I - D_\phi F(\nu,\phi)^{-1}[D_\phi F(\nu,\phi) - D_\phi F(\nu,\phi^H)]\big)^{-1} D_\phi F(\nu,\phi)^{-1}\Big\|_{X;X} \\ \leq 2\|D_\phi F(\nu,\phi)^{-1}\|_{X;X}.$$

Next, we will show (4). Together with the identical equation

$$D_\phi F(\nu,\phi^H) - D_\phi F_h(\nu,\phi^H) = (S - S_h)D_\phi G(\nu,\phi^H)$$

the hypothesis (H4) and (3) imply that there exists $h_*(\leq \bar{h})$ such that for all $h < h_*$,

$$\|D_\phi F(\nu,\phi^H)^{-1}[D_\phi F(\nu,\phi^H) - D_\phi F_h(\nu,\phi^H)]\|_{X;X} \leq 1/2.$$

Considering

$$D_\phi F_h(\nu,\phi^H) = D_\phi F(\nu,\phi^H)(I - D_\phi F(\nu,\phi^H)^{-1}[D_\phi F(\nu,\phi^H) - D_\phi F_h(\nu,\phi^H)]),$$

we see that $D_\phi F_h(\nu,\phi^H)$ is an isomorphism of X and

$$\|D_\phi F_h(\nu,\phi^H)^{-1}\|_{X;X} \leq 2\|D_\phi F(\nu,\phi^H)^{-1}\|_{X;X} \leq 4\|D_\phi F(\nu,\phi)^{-1}\|_{X;X}.$$

□

As it is well-known, the Newton–Raphson method shows quadratic convergence behaviors. Indeed, we prove that Xu's two-grid algorithm in the unified setting has quadratic convergence in the following sense:

Theorem 2.3 *Assume that (H1)–(H4) hold. For $H < h^*$ and $h < h_*$ where h^* and h_* are given in Lemma 2.2, if $\phi^H(\nu)$ and $\phi_h(\nu)$ are solutions obtained from* ***Algorithm 1*** *and if $\phi^h(\nu)$ is a nonsingular solutions of (2), then there exists a positive constant C independent of mesh sizes h and H such that*

$$\|\phi^h(\nu) - \phi_h(\nu)\|_X \le C\|\phi^h(\nu) - \phi^H(\nu)\|_X^2. \tag{5}$$

Moreover, we have

$$\begin{aligned}\|\phi(\nu) - \phi_h(\nu)\|_X \le C\big(&\|(S - S_h)G(\nu, \phi(\nu))\|_X + \|(S - S_h)G(\nu, \phi(\nu))\|_X^2 \\ &+ \|(S - S_H)G(\nu, \phi(\nu))\|_X^2\big).\end{aligned} \tag{6}$$

Proof For simplicity, ν is dropped from the functions $\phi(\nu)$, $\phi_h(\nu)$ and $\phi^H(\nu)$. From Lemma 2.2 and **Algorithm 1**, we get

$$\begin{aligned}\phi_h - \phi^h &= \phi^H - \phi^h + D_\phi F_h(\nu, \phi^H)^{-1}(F_h(\nu, \phi^h) - F_h(\nu, \phi^H)) \\ &= D_\phi F_h(\nu, \phi^H)^{-1}\Big[F_h(\nu, \phi^h) - F_h(\nu, \phi^H) - D_\phi F_h(\nu, \phi^H)(\phi^h - \phi^H)\Big].\end{aligned}$$

The identity

$$F_h(\nu, \phi^h) - F_h(\nu, \phi^H) = \int_0^1 D_\phi F_h(\nu, \phi^H + t(\phi^h - \phi^H))(\phi^h - \phi^H)\, dt$$

yields

$$\begin{aligned}\phi_h - \phi^h &= D_\phi F_h(\nu, \phi^H)^{-1}\int_0^1 \Big[D_\phi F_h(\nu, \phi^H + t(\phi^h - \phi^H)) - D_\phi F_h(\nu, \phi^H)\Big] \\ &\quad (\phi^h - \phi^H)\, dt \\ &= D_\phi F_h(\nu, \phi^H)^{-1}\int_0^1 S_h\Big[D_\phi G(\nu, \phi^H + t(\phi^h - \phi^H)) - D_\phi G(\nu, \phi^H)\Big] \\ &\quad (\phi^h - \phi^H)\, dt.\end{aligned}$$

Using (H3) and the uniform-boundedness theorem we get

$$\|S_h\|_{Y;X} \le C. \tag{7}$$

Therefore, it follows from (4) and (H1) that

$$\|\phi_h - \phi^h\|_X \le C\|\phi^H - \phi^h\|_X^2.$$

Finally, let us prove the estimate (6). First, we have by the triangle inequality

$$\|\phi - \phi_h\|_X \le \|\phi - \phi^h\|_X + \|\phi^h - \phi_h\|_X.$$

Then, the result follows from Theorem 2.1 and (5). □

Remark 2.4 Theorem 2.3 implies that convergence results in terms of the fine scale h is retained under proper relation between fine grid h and coarse grid H. Existing results [8] show that the relation is typically given by $h = \mathcal{O}(H^\alpha)$ for some α with $1 < \alpha \leq 2$. It shows the correct scaling is $h = \mathcal{O}(H^2)$ as long as $\|(S - S_h)G(\nu, \phi(\nu))\|_X = \mathcal{O}(h^\beta)$ for some positive β. Moreover, convergence results in terms of the fine scale h is retained under two-level algorithm.

Algorithm 2: Two-grid algorithm with fine mesh correction
Step 1: Solve nonlinear system on coarse mesh
Find $\phi^H(\nu) \in X_H$ such that $F_H(\nu, \phi^H(\nu)) = 0$.
Step 2: Update on fine mesh with one Newton iteration
Find $\phi_h(\nu) \in X_h$ such that $D_\phi F_h(\nu, \phi^H(\nu))(\phi_h(\nu) - \phi^H(\nu)) = -F_h(\nu, \phi^H(\nu))$.
Step 3: Correct on *fine* mesh
Find $e_h^*(\nu) \in X_h$ such that $D_\phi F_h(\nu, \phi^H(\nu))e_h^*(\nu) = -\frac{1}{2}D_\phi^2 F_h(\nu, \phi^H(\nu))(\phi_h(\nu) - \phi^H(\nu))^2$
and set $\phi_h^*(\nu) := \phi_h(\nu) + e_h^*(\nu)$.

Note that linear systems in Steps 2 and 3 share the same stiffness matrix, which can be beneficial in numerical implementation as well as error analysis. In practice, Step 3 may not be necessary, but this could be a viable option for two-grid strategies.

Theorem 2.5 *Suppose that the assumptions of Theorem 2.3 hold. And assume that G is a $\mathcal{C}^3$-mapping and $D_\phi^3 G$ is bounded on all bounded subsets of $\Lambda \times X$. If $\phi^H(\nu)$, $\phi_h(\nu)$ and $\phi_h^*(\nu)$ are solutions obtained from **Algorithm 2** and if $\phi^h(\nu)$ is a nonsingular solution of (2), then there exists a positive constant C independent of mesh sizes h and H such that*

$$\|\phi^h(\nu) - \phi_h^*(\nu)\|_X \leq C\|\phi^h(\nu) - \phi^H(\nu)\|_X^3. \tag{8}$$

Moreover, we have

$$\begin{aligned}\|\phi(\nu) - \phi_h^*(\nu)\|_X \leq C\big(&\|(S - S_h)G(\nu, \phi(\nu))\|_X \\ &+ \|(S - S_h)G(\nu, \phi(\nu))\|_X^3 + \|(S - S_H)G(\nu, \phi(\nu))\|_X^3\big).\end{aligned} \tag{9}$$

Proof For brevity, we omit ν from the functions $\phi(\nu)$, $\phi_h(\nu)$, $\phi_h^*(\nu)$ and $\phi^H(\nu)$. By the Taylor expansion, we have

$$\begin{aligned}0 = F_h(\nu, \phi^h) = F_h(\nu, \phi^H) &+ D_\phi F_h(\nu, \phi^H)(\phi^h - \phi^H) \\ &+ \frac{1}{2}D_\phi^2 F_h(\nu, \phi^H)(\phi^h - \phi^H)^2 + M(\phi^h - \phi^H)^3,\end{aligned}$$

which yields with the definitions of ϕ_h and e_h^* that

$$\begin{aligned} D_\phi F_h(\nu, \phi^H)(\phi_h^* - \phi^h) &= D_\phi F_h(\nu, \phi^H)(e_h + e_h^* + \phi^H - \phi^h) \\ &= \frac{1}{2} D_\phi^2 F_h(\nu, \phi^H)((\phi^h - \phi^H)^2 - e_h^2) + M(\phi^h - \phi^H)^3, \end{aligned}$$

where $e_h = \phi_h - \phi^H$ and the operator $M \in \mathscr{L}_3(X; Y)$ is defined by $M = \int_0^1 \frac{1}{2}(1 - t)^2 D_\phi^3 F_h(\nu, \phi^H + t(\phi^h - \phi^H))dt$. So, we get

$$\begin{aligned} \phi_h^* - \phi^h = &\frac{1}{2} D_\phi F_h(\nu, \phi^H)^{-1} D_\phi^2 F_h(\nu, \phi^H)(\phi^h - \phi_h)(\phi^h - \phi^H + e_h) \\ &+ D_\phi F_h(\nu, \phi^H)^{-1} M(\phi^h - \phi^H)^3. \end{aligned}$$

Hence

$$\begin{aligned} \|\phi_h^* - \phi^h\|_X \le &\frac{1}{2} \|D_\phi F_h(\nu, \phi^H)^{-1}\|_{X;X} \|D_\phi^2 F_h(\nu, \phi^H)\|_{X^2;X} \|\phi^h \\ &- \phi_h\|_X \|\phi^h - \phi^H + e_h\|_X \\ &+ C\|D_\phi F_h(\nu, \phi^H)^{-1}\|_{X;X} \|\phi^h - \phi^H\|_X^3. \end{aligned}$$

Lemma 2.2, the identity $D_\phi^2 F_h(\nu, \phi^H) = S_h D_\phi^2 G(\nu, \phi^H)$, (7), (5) and the triangle inequality imply that

$$\begin{aligned} \|\phi_h^* - \phi^h\|_X &\le C\|\phi^h - \phi^H\|_X^2 \big(\|\phi^h - \phi^H\|_X \\ &\quad + \|\phi^h - \phi_h\|_X + \|\phi^h - \phi^H\|_X\big) + C\|\phi^h - \phi^H\|_X^3 \\ &\le C\|\phi^h - \phi^H\|_X^3. \end{aligned}$$

Next, to prove the estimate (9), we first use the triangle inequality

$$\|\phi - \phi_h^*\|_X \le \|\phi - \phi^h\|_X + \|\phi^h - \phi_h^*\|_X.$$

Then, the result follows from Theorem 2.1, (8), and the inequality $(a+b)^3 \le 4(a^3+b^3)$ for $a,\ b > 0$. □

Remark 2.6 Theorem 2.5 indicates that $h = \mathcal{O}(H^3)$ is the correct scaling.

Now, in order to estimate the error $\|\phi(\nu) - \phi^h(\nu)\|_E$ for a Banach space E such that $X \hookrightarrow E$, we introduce the theory developed in [11]. We additionally assume that there exists a Banach space V with $V \hookrightarrow X$ such that the following property holds:

$$\begin{cases} \text{for all } v \in V, \text{ the operator } D_\phi G(\lambda, v) \text{ may be extended as a linear operator of } \mathscr{L}(E; Y), \\ \text{the mapping } v \to D_\phi G(\lambda, v) \text{ being continuous from } V \text{ into } \mathscr{L}(E; Y). \end{cases} \tag{10}$$

Hence, for $v \in V$, both $D_\phi F(\lambda, v)$ and $D_\phi F_h(\lambda, v)$ may be extended as operators of $\mathscr{L}(E; E)$. Next, we suppose that

$$\lim_{h \to 0} \|S - S_h\|_{Y;E} = 0. \tag{11}$$

Then we have the following result.

Theorem 2.7 *We retain the hypotheses of Theorem 2.1 together with (10) and (11). Assume in addition that:*

$$for\ each\ \lambda \in \Lambda, \phi(\lambda) \in V\ and\ the\ function\ \lambda \to \phi(\lambda) \in C^0(\Lambda; V), \tag{12}$$

$$for\ each\ \lambda \in \Lambda, D_\phi F(\lambda, \phi(\lambda))\ is\ an\ isomorphism\ of\ E. \tag{13}$$

Then, for $h \leq h_1$ small enough, exists a constant $C > 0$, independent of h and λ, such that

$$\|\phi(\lambda) - \phi^h(\lambda)\|_E \leq C\big(\|(S - S_h)G(\lambda, \phi(\lambda))\|_E + \|\phi(\lambda) - \phi^h(\lambda)\|_X^2.\big) \tag{14}$$

Proof With a simple modification, the proof follows immediately from Theorem 3.5 of Chapter IV in [11]. □

Let $\phi^H(\nu)$, $\phi_h(\nu)$ and $\phi_h^*(\nu)$ be solutions obtained from **Algorithm 2** and let $\phi^h(\nu)$ be a nonsingular solutions of (2). Then together with conditions (10)–(13), the following results can be proved in a similar way to proofs of Lemma 2.2, Theorem 2.3 and Theorem 2.5.

(R1) $D_\phi F_h(\nu, \phi^H(\nu))$ are isomorphisms of E with

$$\|D_\phi F_h(\nu, \phi^H(\nu))^{-1}\|_{E;E} \leq 4\|D_\phi F(\nu, \phi(\nu))^{-1}\|_{E;E}.$$

(R2) There exists a positive constant C independent of mesh sizes h and H such that

$$\|\phi^h(\nu) - \phi_h(\nu)\|_E \leq C\|\phi^h(\nu) - \phi^H(\nu)\|_E^2.$$

Moreover, we have

$$\begin{aligned}\|\phi(\nu) - \phi_h(\nu)\|_E \leq & C\big(\|(S - S_h)G(\nu, \phi(\nu))\|_E + \|\phi(\nu) - \phi^h(\nu)\|_X^2 \\ & + \|(S - S_h)G(\nu, \phi(\nu))\|_E^2 + \|\phi(\nu) - \phi^h(\nu)\|_X^4 \\ & + \|(S - S_H)G(\nu, \phi(\nu))\|_E^2 + \|\phi(\nu) - \phi^H(\nu)\|_X^4\big).\end{aligned}$$

(R3) There exists a positive constant C independent of mesh sizes h and H such that

$$\|\phi^h(\nu) - \phi_h^*(\nu)\|_E \leq C\|\phi^h(\nu) - \phi^H(\nu)\|_E^3.$$

Moreover, we have

$$\begin{aligned}\|\phi(\nu) - \phi_h^*(\nu)\|_E \leq & C\big(\|(S - S_h)G(\nu, \phi(\nu))\|_E + \|\phi(\nu) - \phi^h(\nu)\|_X^2 \\ & + \|(S - S_h)G(\nu, \phi(\nu))\|_E^3 + \|\phi(\nu) - \phi^h(\nu)\|_X^6 \\ & + \|(S - S_H)G(\nu, \phi(\nu))\|_E^3 + \|\phi(\nu) - \phi^H(\nu)\|_X^6\big).\end{aligned}$$

3 Application for Algorithms 1 and 2

In order to illustrate our theory, we present three examples of nonlinear problems; a semilinear second-order elliptic problem, elliptic problems with gradient nonlinearities and the Navier–Stokes equations, and discuss their finite element approximations using two-grid algorithms. We assume that Ω is smooth in $\mathbb{R}^d$ and $\mathcal{T}_h$ is a family of shape regular triangulations of $\overline{\Omega}$ by triangles (d=2) and tetrahedra (d=3) T of diameter h with at most one curved side.

3.1 Semilinear problems

We consider the semilinear boundary value problems of the form

$$\begin{aligned} -\triangle u + G(\lambda, u) &= 0 \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \partial\Omega, \end{aligned} \tag{15}$$

where $G : \mathcal{D}(G) \subset \mathbb{R} \times H_0^1(\Omega) \to L^2(\Omega)$ is given mapping. This problem has been treated in [5] for $d = 2$. Multiplying by test functions and integrating by parts we obtain the following weak formulation of problem (15): find $u \in H_0^1(\Omega)$ such that

$$\int_\Omega \nabla u \cdot \nabla v dx + \int_\Omega G(\lambda, u) v dx = 0, \quad \forall v \in H_0^1(\Omega). \tag{16}$$

We define the continuous linear operator $S : f \in L^2(\Omega) \to Sf = w \in H_0^1(\Omega) \cap H^2(\Omega)$ which maps f onto the solution w of the following problem: given a function $f \in L^2(\Omega)$, find the unique solution $w \in H_0^1(\Omega) \cap H^2(\Omega)$ such that

$$\int_\Omega \nabla w \cdot \nabla v dx = \int_\Omega f v dx, \quad \forall v \in H_0^1(\Omega).$$

Introducing the operator S, problem (16) can be written in the equivalent way

$$F(\lambda, u) = u + SG(\lambda, u) = 0. \tag{17}$$

In order to approximate the solution of problem (17), given a finite-dimensional space $X_h \subset H_0^1(\Omega)$, we define the linear operator $S_h : f \in L^2(\Omega) \to S_h f = w_h \in X_h$ such that

$$\int_\Omega \nabla w_h \cdot \nabla v_h dx = \int_\Omega f v_h dx, \quad \forall v_h \in X_h.$$

Then the approximation of nonlinear problem (17) is to find $(\lambda, u^h) \in (\mathbb{R} \times X_h) \cap \mathcal{D}(G)$ such that

$$F_h(\lambda, u^h) = u^h + S_h G(\lambda, u^h) = 0. \tag{18}$$

We analyze the problem (15) with $G(\lambda, u) := G(u) = u^3 - f$ for a given function $f \in L^2(\Omega)$. In order to apply Theorem 2.1, we take $X := H_0^1(\Omega)$ and $Y := Z := L^2(\Omega)$. Since the mapping G is C^∞ and D^2G is bounded on all bounded subsets of X, the hypothesis (H1) holds. And since $H^1(\Omega)$ is continuously imbedded in $L^6(\Omega)$ for $d \leq 3$ by Sobolev imbedding theorem, the function u^3 belongs to $L^2(\Omega)$, i.e., $G(u) \in Y$ and $DG(u) \in \mathscr{L}(X; Z)$ for all $u \in X$. Hence, the hypothesis (H2) holds. Let $X_h \subset H_0^1(\Omega)$ be a conforming finite element space of piecewise polynomials of degree $k \leq 1$ defined on the triangulation $\mathcal{T}_h$. It is well-known that for $f \in L^2(\Omega)$,

$$\|(S - S_h)f\|_{r,2} \leq Ch^{2-r}\|Sf\|_{2,2} \leq Ch^{2-r}\|f\|_{0,2}, \quad r = 0, 1, \tag{19}$$

where $\|\cdot\|_{1,2} := \|\cdot\|_{H^1(\Omega)}$ and $\|\cdot\|_{0,2} := \|\cdot\|_{L^2(\Omega)}$, which yields that the hypotheses (H3)–(H4) hold. From Theorem 3.1 in Section 3 of [5], we have a unique nonsingular solution $u \in H_0^1(\Omega) \cap H^2(\Omega)$ of the problem (15). From Theorem 2.1, there exists a unique nonsingular solution u^h of (18) such that $\|u - u^h\|_X \leq Ch$.

Now, we want to derive an error bound for $\|u - u^h\|_{0,2}$. We take $V := H_0^1(\Omega) \cap H^2(\Omega)$ and $E := L^2(\Omega)$, then $V \hookrightarrow X \hookrightarrow E$. For $v \in V$ and $w \in E$, $DG(v)w \in Y$ by Sobolev imbedding theorem, $H^2(\Omega) \hookrightarrow C^0(\Omega)$ for $d \leq 3$, and the mapping $v \to DG(v)$ is obviously continuous from V into $\mathscr{L}(E; Y)$, i.e., (10) holds. From (19), the equation (11) holds and the assumptions (12)–(13) obviously hold. By Theorem 2.7, there exists a unique nonsingular solution u^h of (18) such that $\|u - u^h\|_E \leq Ch^2$.

For the problem (15) with $G(\lambda, u) = u^3 - f$, **Algorithm 1** reads

Step 1. Find $u^H \in X_H$ such that

$$\int_\Omega \nabla u^H \cdot \nabla v^H dx + \int_\Omega (u^H)^3 v^H dx = \int_\Omega f v^H dx, \quad \forall v^H \in X_H.$$

Step 2. Find $u_h \in X_h$ such that

$$\int_\Omega \nabla u_h \cdot \nabla v_h dx + 3\int_\Omega (u^H)^2 u_h v_h dx = \int_\Omega \big(2(u^H)^3 v_h + f v_h\big) dx, \quad \forall v_h \in X_h.$$

From Theorem 2.3 and (R2), we have error bound for Algorithm 1:

$$\|u - u_h\|_{r,2} \leq C(h^{2-r} + H^{2(2-r)}), \quad r = 0, 1.$$

If we perform an additional step after Algorithm 1, we arrive at **Algorithm 2**:

Step 3. Find $e_h^* \in X_h$ such that

$$\int_\Omega \nabla e_h^* \cdot \nabla v_h dx + 3\int_\Omega (u^H)^2 e_h^* v_h dx = -3\int_\Omega u^H (u_h - u^H)^2 v_h dx, \quad \forall v_h \in X_h,$$

and set $u_h^* = u_h + e_h^*$.

The error bound for Algorithm 2 can be obtained by Theorem 2.5 and (R3). Indeed, since the mapping D^3G is bounded on all bounded subsets of $\Lambda \times X$, we have for $r = 0, 1$,

$$\|u - u_h^*\|_{r,2} \leq C(h^{2-r} + H^{3(2-r)}).$$

3.2 Elliptic problems with gradient nonlinearities

Consider a mixed finite element approximation of the elliptic problem with gradient nonlinearities ([13,18]):

$$\begin{aligned} -\mathrm{div}(a(x)\nabla p) + \ b(x, p, \nabla p) = 0 \ \text{ for } \ x \in \Omega, \\ p(x) = -g \ \text{ for } \ x \in \partial\Omega, \end{aligned} \tag{20}$$

where $a(x)$ is symmetric positive definite such that $0 < \lambda I \leq a(x) \leq \Lambda I$. Assume that a is Lipschitz continuous and the function $b : \bar{\Omega} \times \mathbb{R} \times \mathbb{R}^d \to \mathbb{R}$, $d = 2$ or 3, satisfies that $b(x, y, z)$ is $C^{1,1}$-class with respect to $y \in \mathbb{R}$ and $z \in \mathbb{R}^d$. We shall assume that (20) has a unique solution $p \in H^s(\Omega)$, where $s = 2 + \epsilon$ (d=2) and $s = \frac{5}{2} + \epsilon$ (d=3) with $0 < \epsilon < 1$.

The problem (20) is equivalent to seeking $(p, \boldsymbol{u})$ such that

$$\begin{aligned} a^{-1}\boldsymbol{u} + \nabla p = 0 \quad \text{in } \Omega, \\ \mathrm{div}\boldsymbol{u} + c(p, \boldsymbol{u}) = 0 \quad \text{in } \Omega, \\ p = -g \quad \text{on } \partial\Omega, \end{aligned} \tag{21}$$

where $c(p, \boldsymbol{u}) := b(x, p, \nabla p) = b(x, p, -a^{-1}\boldsymbol{u})$. Let $\boldsymbol{W} := H(\mathrm{div}; \Omega)$. Multiplying by test functions and integrating by parts we obtain the following weak formulation of problem (21): find $(p, \boldsymbol{u}) \in L^2(\Omega) \times \boldsymbol{W}$ such that

$$\begin{aligned} (a^{-1}\boldsymbol{u}, \ \boldsymbol{v}) - (p, \mathrm{div}\boldsymbol{v}) = < g, \boldsymbol{v} \cdot \boldsymbol{n} >, \quad \forall \boldsymbol{v} \in \boldsymbol{W}, \\ (\mathrm{div}\boldsymbol{u}, q) + (c(p, \boldsymbol{u}), q) = 0, \quad \forall q \in L^2(\Omega), \end{aligned} \tag{22}$$

where $(\cdot, \cdot)$ denotes the inner product in $L^2(\Omega)$ and $< \cdot, \cdot >$ that in $L^2(\partial\Omega)$.

Associated with $\mathcal{T}_h$, we take the Raviart–Thomas subspaces $\boldsymbol{W}_h \subset \boldsymbol{W}$ and $Q_h \subset L^2(\Omega)$ of index $k \geq 0$ given as follows:

$$\begin{aligned} \boldsymbol{W}_h := \{\boldsymbol{v} \in H(\mathrm{div}; \Omega) : \forall T \in \mathcal{T}_h, \ \ \boldsymbol{v}|_T \in RT_k(T)\}, \\ Q_h := \{q \in L^2(\Omega) : \forall T \in \mathcal{T}_h, \ \ q|_T \in P_k(T)\}, \end{aligned} \tag{23}$$

where $RT_k(T)$ is Raviart–Thomas element over T and $P_k(T)$ is the space of polynomials of degree less than or equal k. The mixed approximation of (22) is defined by

finding $(p^h, \boldsymbol{u}^h) \in Q_h \times \boldsymbol{W}_h$ such that

$$\begin{aligned} (a^{-1}\boldsymbol{u}^h, \boldsymbol{v}) - (p^h, \mathrm{div}\boldsymbol{v}) &= < g, \boldsymbol{v}\cdot\boldsymbol{n} >, \quad \forall \boldsymbol{v} \in \boldsymbol{W}_h, \\ (\mathrm{div}\boldsymbol{u}^h, q) + (c(p^h, \boldsymbol{u}^h), q) &= 0, \quad \forall q \in Q_h. \end{aligned} \tag{24}$$

We introduce the linear solution operator

$$\begin{aligned} S : L^2(\Omega) \times H^{3/2}(\partial\Omega) &\longrightarrow L^2(\Omega) \times \boldsymbol{W} \\ (f^*, g^*) &\mapsto S(f^*, g^*) = (\phi, \boldsymbol{\psi}) \end{aligned}$$

defined by the solution of

$$\begin{aligned} (a^{-1}\boldsymbol{\psi}, \boldsymbol{v}) - (\phi, \mathrm{div}\boldsymbol{v}) &= < g^*, \boldsymbol{v}\cdot\boldsymbol{n} >, \quad \forall \boldsymbol{v} \in H(\mathrm{div}; \Omega), \\ (\mathrm{div}\boldsymbol{\psi}, q) &= (f^*, q), \quad \forall q \in L^2(\Omega). \end{aligned} \tag{25}$$

It is well-known [3] that for any $(f^*, g^*) \in L^2(\Omega) \times H^{3/2}(\partial\Omega)$, the problem (25) has a unique solution $\psi = (\phi, \boldsymbol{\psi}) \in H^2(\Omega) \times [H^1(\Omega)]^2$ such that

$$\|\phi\|_{2,2} + \|\boldsymbol{\psi}\|_{1,2} \le C\big(\|f^*\|_{0,2} + |g^*|_{\frac{3}{2},2}\big)$$

Let us define the discrete solution operator

$$\begin{aligned} S_h : L^2(\Omega) \times H^{3/2}(\partial\Omega) &\longrightarrow X_h \\ (f^*, g^*) &\mapsto S_h(f^*, g^*) = (\phi_h, \boldsymbol{\psi}_h) \end{aligned}$$

by

$$\begin{aligned} (a^{-1}\boldsymbol{\psi}_h, \boldsymbol{v}) - (\phi_h, \mathrm{div}\boldsymbol{v}) &= < g^*, \boldsymbol{v}\cdot\boldsymbol{n} >_{\partial\Omega}, \quad \forall \boldsymbol{v} \in \boldsymbol{W}_h, \\ (\mathrm{div}\boldsymbol{\psi}_h, q) &= (f^*, q), \quad \forall q \in Q_h. \end{aligned} \tag{26}$$

This is the discrete problem corresponding to the continuous one (25). It is a well-known result that this problem is well-posed and has an acceptable error estimate.

In order to apply Theorem 2.1, we take $X := L^m(\Omega) \times L^m(\Omega)^d$, $Y := L^2(\Omega) \times H^{3/2}(\partial\Omega)$ and $Z := L^2(\Omega) \times \{0\}$. We define the mapping

$$\begin{aligned} G : X &\to Y \\ v = (q, \boldsymbol{v}) &\mapsto G(v) \end{aligned}$$

by $G(v) := (c(q, \boldsymbol{v}), -g)$. Recall that $c(q, \boldsymbol{v}) = b(q, -a^{-1}\boldsymbol{v})$. We will determine later the value of m satisfying $G(v) \in Y$ and $DG(v)w \in Z$ for all $v, w \in X$. From the assumption of the function b, the mapping G is C^2 and D^2G is bounded on all bounded subsets of X. Hence, the hypotheses (H1)–(H2) hold.

Clearly, a pair $u := (p, \boldsymbol{u})$ satisfies the equation

$$F(u) := u + SG(u) = 0 \tag{27}$$

if and only if the function $u = (p, \boldsymbol{u}) \in X$ satisfies the equation (22). Moreover, F is Fréchet differentiable on X and for the unique solution $u = (p, \boldsymbol{u}) \in X$ of (21), $DF(u) = I + SDG(v)$ is an isomorphism of X (see [13]). We can construct a family $\{F_h\}_{0<h<1}$ of mappings, $F_h : X \to X$, by for all $v \in X$

$$F_h(v) := v + S_h G(v).$$

Thus, an equivalent form of problem (24) consists in finding the solution $u^h = (p^h, \boldsymbol{u}^h) \in X_h$ satisfying

$$F_h(u^h) = u^h + S_h G(u^h) = 0, \tag{28}$$

where $G(u^h) = (c(p^h, \boldsymbol{u}^h), -g)$.

We can prove that (cf. [13,18]) if the solution $(\phi, \boldsymbol{\psi})$ of problem (25) belongs to $H^{k+2}(\Omega) \times [H^{k+1}(\Omega)]^2$ for $(f^*, g^*) \in H^k(\Omega) \times H^{k+3/2}(\partial\Omega)$, then there exists a constant $C > 0$ independent of h such that for integer $k \geq 0$ and $\frac{d}{m} - \frac{d-2}{2} \leq s \leq k + \frac{d}{m} - \frac{d-2}{2}$,

$$\begin{aligned}
\|(S - S_h)(f^*, g^*)\|_X &\leq \|\boldsymbol{\psi} - \boldsymbol{\psi}_h\|_{0,m} + \|\phi - \phi_h\|_{0,m} \\
&\leq Ch^s \left(\|\boldsymbol{\psi}\|_{s-\frac{d}{m}+\frac{d}{2},2} + \|\phi\|_{s-\frac{d}{m}+\frac{d}{2},2} \right) \\
&\leq Ch^s \left(\|f^*\|_{s-\frac{d}{m}+\frac{d-2}{2},2} + |g^*|_{s-\frac{d}{m}+\frac{d+1}{2},2} \right).
\end{aligned}$$

Particularly, for $k = 0$, we take $s = \frac{2}{m}, m < \infty$, for $d = 2$ and $s = \frac{3}{m} - \frac{1}{2}, m < 6$, for $d = 3$. The hypotheses (H3)–(H4) hold by above error estimates and hence from Theorem 2.1, there exists a unique nonsingular solution $u^h \in X_h$ of (37) such that $\|u - u^h\|_X \leq Ch^s$.

For the problem (20), **Algorithm 1** (Steps 1 & 2) and **Algorithm 2** (Steps 1 to 3) read

Step 1. Find $u^H = (p^H, \boldsymbol{u}^H) \in X_H$ such that $\forall v^H = (q^H, \boldsymbol{v}^H) \in X_H$,

$$\begin{aligned}
\int_\Omega a^{-1}\boldsymbol{u}^H \cdot \boldsymbol{v}^H dx - \int_\Omega p^H \mathrm{div}\boldsymbol{v}^H dx &= \int_{\partial\Omega} g\boldsymbol{v}^H \cdot \boldsymbol{n} ds, \\
\int_\Omega \mathrm{div}\boldsymbol{u}^H q^H dx + \int_\Omega c(p^H, \boldsymbol{u}^H) q^H dx &= 0.
\end{aligned}$$

Step 2. Find $u_h = (p_h, \boldsymbol{u}_h) \in X_h$ such that $\forall v_h = (q_h, \boldsymbol{v}_h) \in X_h$,

$$\begin{aligned}
&\int_\Omega a^{-1}\boldsymbol{u}_h \cdot \boldsymbol{v}_h dx - \int_\Omega p_h \mathrm{div}\boldsymbol{v}_h dx = \int_{\partial\Omega} g\boldsymbol{v}_h \cdot \boldsymbol{n} ds, \\
&\int_\Omega \mathrm{div}\boldsymbol{u}_h q_h dx + \int_\Omega \Big(c_q(p^H, \boldsymbol{u}^H) p_h + c_{\boldsymbol{v}}(p^H, \boldsymbol{u}^H) \cdot \boldsymbol{u}_h \Big) q_h dx \\
&= \int_\Omega \Big(c_q(p^H, \boldsymbol{u}^H) p^H + c_{\boldsymbol{v}}(p^H, \boldsymbol{u}^H) \cdot \boldsymbol{u}^H - c(p^H, \boldsymbol{u}^H) \Big) q_h dx.
\end{aligned}$$

Step 3. Find $e_h^* = ({p_h}^*, {\boldsymbol{u}_h}^*) \in X_h$ such that $\forall v_h = (q_h, \boldsymbol{v}_h) \in X_h$,

$$\begin{aligned}
&\int_\Omega a^{-1}{\boldsymbol{u}_h}^* \cdot \boldsymbol{v}_h dx - \int_\Omega {p_h}^* \mathrm{div}\boldsymbol{v}_h dx = 0,\\
&\int_\Omega \mathrm{div}{\boldsymbol{u}_h}^* q_h dx + \int_\Omega \Big(c_q(p^H, \boldsymbol{u}^H){p_h}^* + c_{\boldsymbol{v}}(p^H, \boldsymbol{u}^H)\cdot {\boldsymbol{u}_h}^*\Big) q_h dx\\
&= \int_\Omega \Big(-\frac{1}{2}c_{qq}(p^H, \boldsymbol{u}^H)(p_h - p^H)^2 - c_{q\boldsymbol{v}}(p^H, \boldsymbol{u}^H)(p_h - p^H)\cdot(\boldsymbol{u}_h - \boldsymbol{u}^H)\\
&-\frac{1}{2}c_{\boldsymbol{vv}}(p^H, \boldsymbol{u}^H)(\boldsymbol{u}_h - \boldsymbol{u}^H)^2\Big) q_h dx,
\end{aligned}$$

and set $u_h^* = u_h + e_h^*$.

It follow from Theorem 2.3 that $\|u - u_h\|_X \le C(h^s + H^{2s})$. Further, if D^3G is bounded on all bounded subsets of $\Lambda \times X$, then by Theorem 2.5 we get $\|u - u_h^*\|_X \le C(h^s + H^{3s})$. For example, if $b(x, p, \nabla p) = b_1 p^2 + b_2|\nabla p|^2 - b_0(x)$ for some constants b_1, b_2 and functions $b_0(x)$, then all conditions on the function b hold together with $m \ge 4$ for $d = 2$ and $4 \le m < 6$ for $d = 3$. Indeed, in this case, D^2G and D^3G are bounded on all bounded subsets of X. Lastly, we mention that two-grid mixed methods were studied previously in [6–8,14,20,21] for linear and semilinear problems.

3.3 Navier–Stokes equations

Given the body force $\boldsymbol{f}$, the stationary and incompressible Navier–Stokes equations for the unknown velocity $\boldsymbol{u}$ and the pressure p reads

$$\begin{aligned}
-\nu\triangle\boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} + \nabla p &= \boldsymbol{f} \quad \text{in } \Omega,\\
\nabla\cdot\boldsymbol{u} &= 0 \quad \text{in } \Omega,\\
\boldsymbol{u} &= \boldsymbol{0} \quad \text{on } \partial\Omega,
\end{aligned} \tag{29}$$

where the positive number ν. The compatibility condition read

$$\int_\Omega p\, dx = 0.$$

It was shown in [11] that this problem fits into the BRR framework. But for the reader's convenience we describe briefly how the problem fits into the BRR framework. Rescaling the pressure and the body force by $p/\nu \to p$ and $\boldsymbol{f}/\nu \to \boldsymbol{f}$, respectively, we have

$$\begin{aligned}
-\triangle\boldsymbol{u} + \frac{1}{\nu}(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} + \nabla p &= \boldsymbol{f} \quad \text{in } \Omega,\\
\nabla\cdot\boldsymbol{u} &= 0 \quad \text{in } \Omega,\\
\boldsymbol{u} &= \boldsymbol{0} \quad \text{on } \partial\Omega.
\end{aligned} \tag{30}$$

We introduce the spaces $L_0^2(\Omega) := \{q \in L^2(\Omega) : \int_\Omega q dx = 0\}$, $X := H_0^1(\Omega)^d \times L_0^2(\Omega)$, $Y := H^{-1}(\Omega)^d$, and $Z := L^{\frac{3}{2}}(\Omega)^d$. We know that (30) is equivalent to the variational formulation: given $\boldsymbol{f} \in Y$, find $(\boldsymbol{u}, p) \in H_0^1(\Omega)^d \times L_0^2(\Omega)$ such that

$$\int_\Omega \nabla \boldsymbol{u} : \nabla \boldsymbol{v} dx + \frac{1}{\nu} \int_\Omega (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} \cdot \boldsymbol{v} dx - \int_\Omega p \nabla \cdot \boldsymbol{v} = \int_\Omega \boldsymbol{f} \cdot \boldsymbol{v} dx \quad \forall \boldsymbol{v} \in H_0^1(\Omega)^d,$$
$$\int_\Omega q \nabla \cdot \boldsymbol{u} dx = 0 \quad \forall q \in L_0^2(\Omega). \tag{31}$$

Next we define a linear solution operator S as follows: given $\bar{\boldsymbol{f}} \in Y$, we denote by $(\bar{\boldsymbol{u}}, \bar{p}) = S\bar{\boldsymbol{f}} \in X$ the solution of the Dirichlet problem for the Stokes equations: for all $(\boldsymbol{v}, q) \in X$,

$$\int_\Omega \nabla \bar{\boldsymbol{u}} : \nabla \boldsymbol{v} dx - \int_\Omega \bar{p} \nabla \cdot \boldsymbol{v} dx = \int_\Omega \bar{\boldsymbol{f}} \cdot \boldsymbol{v} dx,$$
$$\int_\Omega q \nabla \cdot \bar{\boldsymbol{u}} dx = 0. \tag{32}$$

Consider a finite dimensional subspace $X_h \subset X$. The variational formulation (31) is discretized by the following problem: find $(\boldsymbol{u}^h, p^h) \in X_h$ such that for all $(\boldsymbol{v}^h, q^h) \in X_h$,

$$\int_\Omega \nabla \boldsymbol{u}^h : \nabla \boldsymbol{v}^h dx + \frac{1}{\nu} \int_\Omega (\boldsymbol{u}^h \cdot \nabla)\boldsymbol{u}^h \cdot \boldsymbol{v}^h dx - \int_\Omega p^h \nabla \cdot \boldsymbol{v}^h = \int_\Omega \boldsymbol{f} \cdot \boldsymbol{v}^h dx,$$
$$\int_\Omega q^h \nabla \cdot \boldsymbol{u}^h dx = 0. \tag{33}$$

The associated discrete solution operator S_h is defined as follows. For a given $\bar{\boldsymbol{f}} \in Y$, we denote by $(\bar{\boldsymbol{u}}_h, \bar{p}_h) = S_h \bar{\boldsymbol{f}} \in X_h$ the approximate solution of the following equations: for all $(\boldsymbol{v}_h, q_h) \in X_h$,

$$\int_\Omega \nabla \bar{\boldsymbol{u}}_h : \nabla \boldsymbol{v}_h dx - \int_\Omega \bar{p}_h \nabla \cdot \boldsymbol{v}_h dx = \int_\Omega \bar{\boldsymbol{f}} \boldsymbol{v}_h dx,$$
$$\int_\Omega q_h \nabla \cdot \bar{\boldsymbol{u}}_h dx = 0. \tag{34}$$

With the data $\boldsymbol{f} \in Y$ we associate a C^∞-mapping $G : \mathbb{R}_+ \times X \to Y$ defined by

$$G(\lambda, \psi) = \lambda(\boldsymbol{v} \cdot \nabla)\boldsymbol{v} - \lambda \boldsymbol{f},$$

where $\psi = (\boldsymbol{v}, q) \in X$ and $\lambda = 1/\nu$. We set

$$F(\lambda, \psi) = \psi + SG(\lambda, \psi) \tag{35}$$

and

$$F_h(\lambda, \psi) = \psi + S_h G(\lambda, \psi). \tag{36}$$

We can prove that the pair $(\boldsymbol{u}, p) \in X$ is a solution of problem (31) if and only if $\phi = (\boldsymbol{u}, p) \in X$ is a solution of $F(\lambda, \phi) = 0$ where $\lambda = 1/\nu$ and that $\phi = (\boldsymbol{u}, p)$ is a nonsingular solution of $F(\lambda, \phi) = 0$ (see, Lemma 3.1–3.2 of Chapter IV in [11]). And an equivalent form of problem (33) consists in finding the solution $\phi^h = (\boldsymbol{u}^h, p^h) \in X_h$ satisfying

$$F_h(\lambda, \phi^h) = \phi^h + S_h G(\lambda, \phi^h) = 0, \tag{37}$$

where $G(\lambda, \phi^h) = \lambda(\boldsymbol{u}^h \cdot \nabla)\boldsymbol{u}^h - \lambda \boldsymbol{f}$. Since for $\psi_i = (\boldsymbol{w}_i, q_i),\ i = 1, 2$

$$D^2_\phi G(\lambda, \phi) \cdot (\psi_1, \psi_2) = \lambda((\boldsymbol{w}_1 \cdot \nabla)\boldsymbol{w}_2 + (\boldsymbol{w}_2 \cdot \nabla)\boldsymbol{w}_1),$$

$D^2_\phi G$ is independent of ϕ. Thus, $D^2_\phi G$ is bounded on all bounded subsets of $\Lambda \times X$. So, the hypothesis (H1) holds. Note that by Sobolev imbedding theorem the mapping $u, v \to u \cdot v$ is a continuous bilinear map from $H^1(\Omega) \times L^2(\Omega)$ into $L^2(\Omega)$ for $d = 2$ and $L^{3/2}(\Omega)$ for $d = 3$, which yields that $D_\phi G(\lambda, \phi) \cdot (\psi) = \lambda(\boldsymbol{v} \cdot \nabla)\boldsymbol{u} + \lambda(\boldsymbol{u} \cdot \nabla)\boldsymbol{v} \in Z$ for all $\phi, \psi \in X$. Hence, the hypothesis (H2) holds. We assume that the space X_h satisfies the usual inf-sup or LBB stability condition for compatibility of the velocity-pressure spaces and the standard approximation properties of order $k \geq 1$. Then, from Theorem 1.8 of Chapter II in [11], we have

$$\lim_{h \to 0} \{|\bar{\boldsymbol{u}} - \bar{\boldsymbol{u}}_h|_{1,2,\Omega} + \|\bar{p} - \bar{p}_h\|_{0,2,\Omega}\} = 0,$$

i.e. the hypothesis (H3) holds:

$$\lim_{h \to 0} \|(S - S_h)\boldsymbol{f}\|_X = 0, \quad \forall \boldsymbol{f} \in Y. \tag{38}$$

Note that $L^r(\Omega)$ is compactly imbedded in $H^{-1}(\Omega)$ for $r > 6/5$. So, $Z = L^{3/2}(\Omega) \hookrightarrow Y = H^{-1}(\Omega)$ with a compact imbedding, which yields together with (38) that (H4) holds. Furthermore, if $\bar{\boldsymbol{f}} \in H^{k-1}(\Omega)^d$, then we have from Theorem 1.8 of Chapter II in [11]

$$\|\bar{\boldsymbol{u}} - \bar{\boldsymbol{u}}_h\|_{1,2,\Omega} + \|\bar{p} - \bar{p}_h\|_{0,2,\Omega} \leq Ch^k(\|\bar{\boldsymbol{u}}\|_{k+1,2,\Omega} + \|\bar{p}\|_{k,2,\Omega}) \leq Ch^k\|\bar{\boldsymbol{f}}\|_{k-1,2,\Omega}.$$

If we take the *Mini* or *Hood–Taylor* finite element space (cf., Theorem 4.1 and Theorem 4.3 of Chapter II in [11]) with $k = 1$ as a finite dimensional space X_h, from Theorem 2.1, for each $\boldsymbol{f} \in L^2(\Omega)^d$, there exists a unique nonsingular solution ϕ^h of (36) such that

$$\|\phi - \phi^h\|_X \leq Ch.$$

For the problem (29), we have **Algorithm 1** (Steps 1 and 2) and **Algorithm 2** (Steps 1–3):

Step 1. Find $\phi^H = (\boldsymbol{u}^H, p^H) \in X_H$ such that $\forall v^H \in X_H$,

$$\int_\Omega \nabla \boldsymbol{u}^H : \nabla \boldsymbol{v}^H dx + \int_\Omega \lambda(\boldsymbol{u}^H \cdot \nabla)\boldsymbol{u}^H \cdot \boldsymbol{v}^H dx - \int_\Omega p^H \nabla \cdot \boldsymbol{v}^H dx = \int_\Omega \lambda \boldsymbol{f} \cdot \boldsymbol{v}^H dx.$$

Step 2. Find $\phi_h = (\boldsymbol{u}_h, p_h) \in X_h$ such that $\forall v_h \in X_h$,

$$\begin{aligned} &\int_\Omega \nabla \boldsymbol{u}_h : \nabla \boldsymbol{v}_h dx + \int_\Omega \lambda(\boldsymbol{u}^H \cdot \nabla)\boldsymbol{u}_h \cdot \boldsymbol{v}_h dx + \int_\Omega \lambda(\boldsymbol{u}_h \cdot \nabla)\boldsymbol{u}^H \cdot \boldsymbol{v}_h dx \\ &- \int_\Omega p_h \nabla \cdot \boldsymbol{v}_h dx \quad = \int_\Omega \lambda\big((\boldsymbol{u}^H \cdot \nabla)\boldsymbol{u}^H \cdot \boldsymbol{v}_h + \boldsymbol{f} \cdot \boldsymbol{v}_h\big) dx. \end{aligned}$$

Step 3. Find $e_h^* = (\boldsymbol{u}_h^*, p_h^*) \in X_h$ such that $\forall v_h \in X_h$,

$$\begin{aligned} &\int_\Omega \nabla \boldsymbol{u}_h^* : \nabla \boldsymbol{v}_h dx + \int_\Omega \lambda(\boldsymbol{u}^H \cdot \nabla)\boldsymbol{u}_h^* \cdot \boldsymbol{v}_h dx + \int_\Omega \lambda(\boldsymbol{u}_h^* \cdot \nabla)\boldsymbol{u}^H \cdot \boldsymbol{v}_h dx \\ &- \int_\Omega p_h^* \nabla \cdot \boldsymbol{v}_h dx \qquad = -\int_\Omega \lambda((\boldsymbol{u}_h - \boldsymbol{u}^H) \cdot \nabla)(\boldsymbol{u}_h - \boldsymbol{u}^H) \cdot \boldsymbol{v}_h dx. \end{aligned}$$

and set $\phi_h^* = \phi_h + e_h^*$.

The error bounds for Algorithms 1 and 2 can be obtained by Theorem 2.3 and Theorem 2.5, respectively. Indeed, for Algorithm 1, since the mapping $D_\phi^2 G$ is bounded on all bounded subsets of $\Lambda \times X$, we have

$$\|\phi - \phi_h\|_X \le C(h + H^2),$$

and for Algorithm 2, since $D_\phi^3 G$ is zero, we have

$$\|\phi - \phi_h^*\|_X \le C(h + H^3).$$

We mention that two-grid methods for the Navier–Stokes equations were previously studied in [9,10,15–17].

4 BRR framework with duality pairing and two-grid methods

In this section, we consider the particular abstract framework developed by Caloz and Rappaz [5] for Galerkin approximation of nonlinear problems, which allows higher order relations between meshes if it is applied to a particular example (see Sect. 5). Further, correction step can be done on coarse mesh instead of on fine mesh in Algorithm 2 (cf. [22]).

Let Y' be the dual space of a real Banach space Y. We assume

$$F : X \to Y' \text{ is a } C^1 \text{ mapping and } \phi \in X \text{ is such that } F(\phi) = 0 \text{ and } DF(\phi) \text{ is an isomorphism from } X \text{ onto } Y'. \tag{39}$$

Note that

$$F(\phi) = 0 \Longleftrightarrow \forall y \in Y, \quad \langle F(\phi), y\rangle_{Y'Y} = 0.$$

Let now $\{X_h\}_{0<h\leq 1}$ be a family of finite-dimensional subspacea of X and $\{Y_h\}_{0<h\leq 1}$ be a family of finite-dimensional subspaces of Y. A Galerkin approximation of the problem to find $\phi \in X$ such that

$$F(\phi) = 0 \tag{40}$$

reads: find $\phi^h \in X_h$ such that

$$\forall y^h \in Y_h, \quad \left\langle F(\phi^h), y^h \right\rangle_{Y'Y} = 0. \tag{41}$$

We define the bilinear form $b_\phi : X \times Y \to \mathbb{R}$ by

$$\forall x \in X, \quad y \in Y, \quad b_\phi(x, y) = \langle DF(\phi)x, y\rangle_{Y'Y}$$

We assume there exists a constant $\beta_h > 0$ such that

$$\inf_{\substack{x\in X_h \\ \|x\|_X=1}} \sup_{\substack{y\in Y_h \\ \|y\|_Y=1}} b_\phi(x, y) \geq \beta_h \tag{42}$$

and

$$\dim X_h = \dim Y_h \text{ for } 0 < h \leq 1. \tag{43}$$

Finally we assume

$$\lim_{h\to 0} \inf_{x^h\in X_h} \beta_h^{-2} \|\phi - x^h\|_X = 0. \tag{44}$$

We state the main result proved in [5].

Theorem 4.1 *Let us suppose that hypotheses* (39) *and* (42)–(44) *are satisfied. Moreover, we assume that DF is Lipschitzian at ϕ, that is there exist $\epsilon_0 > 0$ and $L > 0$ such that for all $\psi \in X$ with $\|\phi - \psi\|_X \leq \epsilon_0$,*

$$\|DF(\phi) - DF(\psi)\|_{X;Y'} \leq L\|\phi - \psi\|_X.$$

Then there exist two constants $h_0 > 0$, $\delta_0 > 0$, and for $h \le h_0$ a unique solution ϕ^h to problem (41) in the closed ball $\overline{B}(\phi, \delta_h)$ with $\delta_h = \delta_0 \beta_h$. Moreover, there exists a constant C independent of h such that

$$\|\phi - \phi^h\|_X \le C\beta_h^{-1} \inf_{x^h \in X_h} \|\phi - x^h\|_X. \tag{45}$$

Lemma 4.2 *If assumptions of Theorem 4.1 hold, there exists a constant $\gamma_h > 0$ such that for any $\psi \in X$ with $\|\phi - \psi\|_X \le \min\{\epsilon_0, \beta_h/(2L)\}$,*

$$\inf_{\substack{x \in X_h \\ \|x\|_X = 1}} \sup_{\substack{y \in Y_h \\ \|y\|_Y = 1}} b_\psi(x, y) \ge \gamma_h$$

Proof From the fact that $\|\phi - \psi\|_X \le \epsilon_0$ and $\|\phi - \psi\|_X \le \beta_h/(2L)$, we get the following inequality

$$\begin{aligned} b_\psi(x, y) &= b_\phi(x, y) + \langle (DF(\psi) - DF(\phi))x, y\rangle_{Y'Y} \\ &\ge b_\phi(x, y) - L\|\phi - \psi\|_X \|x\|_X \|y\|_Y \\ &\ge b_\phi(x, y) - \frac{\beta_h}{2}\|x\|_X \|y\|_Y. \end{aligned}$$

From (42) we conclude with $\gamma_h = \beta_h/2$. □

Algorithm 3

Step 1: Solve nonlinear system on coarse mesh

Find $\phi^H \in X_H$ such that $\langle F(\phi^H), y^H\rangle_{Y'Y} = 0$, $\forall y^H \in Y_H$.

Step 2: Update on fine mesh with one Newton iteration

Find $\phi_h \in X_h$ such that $\langle DF(\phi^H)(\phi_h - \phi^H), y_h\rangle_{Y'Y} = \langle -F(\phi^H), y_h\rangle_{Y'Y}$, $\forall y_h \in Y_h$.

Theorem 4.3 *Under assumptions of Theorem 4.1, if ϕ^H and ϕ_h are solutions obtained from **Algorithm 3** and if ϕ^h is a solution of (41), then there exists a positive constant C independent of mesh sizes h and H such that for sufficiently small H*

$$\|\phi^h - \phi_h\|_X \le C\|\phi^h - \phi^H\|_X^2. \tag{46}$$

Proof From **Step 2** of **Algorithm 3**, we get for all $y \in Y_h$,

$$\begin{aligned} b_{\phi^H}(\phi_h - \phi^h, y) &= \left\langle F(\phi^h) - F(\phi^H) - DF(\phi^H)(\phi^h - \phi^H), y\right\rangle_{Y'Y} \\ &= \left\langle \int_0^1 \left[DF(\phi^H + t(\phi^h - \phi^H)) - DF(\phi^H)\right](\phi^h - \phi^H)\, dt, y\right\rangle_{Y'Y}. \end{aligned}$$

If H is sufficiently small, it follows from (44) and (45) that

$$b_{\phi^H}(\phi_h - \phi^h, y) \le \frac{L}{2}\|\phi^h - \phi^H\|_X^2 \|y\|_Y.$$

The inequality (46) follows from Lemma 4.2. □

Algorithm 4

Step 1: Solve nonlinear system on coarse mesh

Find $\phi^H \in X_H$ such that $\langle F(\phi^H), y^H\rangle_{Y'Y} = 0,\ \forall y^H \in Y_H$.

Step 2: Update on fine mesh with one Newton iteration

Find $\phi_h \in X_h$ such that $\langle DF(\phi^H)(\phi_h - \phi^H), y_h\rangle_{Y'Y} = \langle -F(\phi^H), y_h\rangle_{Y'Y}$, $\forall y_h \in Y_h$.

Step 3: Correct on *fine* mesh

Find $e_h^* \in X_h$ such that $\langle DF(\phi^H)e_h^*, y_h\rangle_{Y'Y} = \langle -\frac{1}{2}D^2F(\phi^H)(\phi_h - \phi^H)^2, y_h\rangle_{Y'Y}$ $\forall y_h \in Y_h$

and set $\phi_h^* := \phi_h + e_h^*$.

With arguments similar to the proofs of Theorem 2.5 and Theorem 4.3, we can obtain the following.

Theorem 4.4 *Suppose that the assumptions of Theorem 4.1 hold. And assume that F is a $\mathcal{C}^3$-mapping and D^3F is bounded on all bounded subsets of $\Lambda \times X$. If ϕ^H, ϕ_h and ϕ_h^* are solutions obtained from **Algorithm 4** and if ϕ^h is a nonsingular solution of (41), then there exists a positive constant C independent of mesh sizes h and H such that*

$$\|\phi^h - \phi_h^*\|_X \le C\|\phi^h - \phi^H\|_X^3.$$

Algorithm 5

Step 1: Solve nonlinear system on coarse mesh

Find $\phi^H \in X_H$ such that $\langle F(\phi^H), y^H\rangle_{Y'Y} = 0,\ \forall y^H \in Y_H$.

Step 2: Update on fine mesh with one Newton iteration

Find $\phi_h \in X_h$ such that $\langle DF(\phi^H)(\phi_h - \phi^H), y_h\rangle_{Y'Y} = \langle -F(\phi^H), y_h\rangle_{Y'Y}$, $\forall y_h \in Y_h$.

Step 3: Correct on *coarse* mesh

Find $e_H^* \in X_H$ such that $\langle DF(\phi^H)e_H^*, y_H\rangle_{Y'Y} = \langle -\frac{1}{2}D^2F(\phi^H)(\phi_h - \phi^H)^2, y_H\rangle_{Y'Y},\ \forall y_H \in Y_H$

and set $\phi_h^* := \phi_h + e_H^*$.

The difference between Algorithm 4 and Algorithm 5 lies in Step 3 where a fine grid correction or a coarse grid correction is performed. In order to prove that Algorithm 5 gives the similar result as in Theorem 4.4, we first define the projection $\Pi_H \in \mathcal{L}(Y; Y_H)$ by

$$b_\phi(x, y - \Pi_H y) = 0, \quad \forall x \in X_H, \quad \forall y \in Y. \tag{47}$$

Under hypotheses (42) and (43), Π_H is well defined. We assume that there exists $\epsilon_1 > 0$ such that for any $\psi \in X$ with $\|\phi - \psi\|_X \le \epsilon_1$,

$$|b_\psi(x, y - \Pi_H y)| \le CH\|x\|_X\|y\|_Y \quad \forall x \in X, \quad \forall y \in Y. \tag{48}$$

Theorem 4.5 *Suppose that the assumptions of Theorem 4.1 hold. And assume that F is a C^3-mapping and D^3F is bounded on all bounded subsets of X. If ϕ^H, ϕ_h and ϕ_h^* are solutions obtained from* ***Algorithm 5*** *and if ϕ^h is a nonsingular solution of (41), then there exists a positive constant C independent of mesh sizes h and H such that*

$$\|\phi^h - \phi_h^*\|_X \le C\left(\|\phi^h - \phi^H\|_X^3 + H\|\phi^h - \phi^H\|_X^2\right). \tag{49}$$

Proof By the definitions of ϕ_h and e_H^* and by the Taylor's expansion, we have for all $y \in Y_h$,

$$\begin{aligned}
&b_{\phi^H}(\phi_h^* - \phi^h, y) = b_{\phi^H}(\phi_h - \phi^h, y) + b_{\phi^H}(e_H^*, \Pi_H y)\\
&= \left\langle F(\phi^h) - F(\phi^H) - DF(\phi^H)(\phi^h - \phi^H) - \frac{1}{2}D^2F(\phi^H)(\phi_h - \phi^H)^2, y\right\rangle_{Y'Y}\\
&\quad + \left\langle \frac{1}{2}D^2F(\phi^H)(\phi_h - \phi^H)^2, y - \Pi_H y\right\rangle_{Y'Y}\\
&= \left\langle \frac{1}{2}D^2F(\phi^H)\left((\phi^h - \phi^H)^2 - (\phi_h - \phi^H)^2\right), y\right\rangle_{Y'Y} + \left\langle M(\phi^h - \phi^H)^3, y\right\rangle_{YY'}\\
&\quad + \left\langle \frac{1}{2}D^2F(\phi^H)(\phi_h - \phi^H)^2, y - \Pi_H y\right\rangle_{Y'Y}.
\end{aligned}$$

where the operator $M = \int_0^1 \frac{1}{2}(1-t)^2 D^3F(\phi^H + t(\phi^h - \phi^H))dt$. By Theorem 4.3

$$\begin{aligned}
\left\langle \frac{1}{2}D^2F(\phi^H)\left((\phi^h - \phi^H)^2 - (\phi_h - \phi^H)^2\right), y\right\rangle_{Y'Y} &\le C\|\phi^h\\
&\quad - \phi_h\|_X\|\phi^h - \phi^H + e_h\|_X\|y\|_Y\\
&\le C\|\phi^h - \phi^H\|_X^3\|y\|_Y.
\end{aligned}$$

If H is sufficiently small, by the assumption that D^3F is bounded on all bounded subsets of X we have

$$\left\langle M(\phi^h - \phi^H)^3, y\right\rangle_{YY'} \le C\|\phi^h - \phi^H\|_X^3\|y\|_Y$$

and by (48) we have

$$\left\langle \frac{1}{2}D^2F(\phi^H)(\phi_h - \phi^H)^2, y - \Pi_H y\right\rangle_{Y'Y} \le CH\|\phi^h - \phi^H\|_X^2\|y\|_Y$$

The inequality (49) follows from Lemma 4.2. □

5 Application for Algorithms 3, 4 and 5

We can apply **Algorithm 3** to the semilinear problems

$$\begin{aligned} -\triangle u + G(u) &= 0 \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \partial\Omega, \end{aligned} \tag{50}$$

where Ω is a bounded open set in $\mathbb{R}^d$ with smooth boundary $\partial\Omega$ and $G : \mathcal{D}(G) \subset H_0^1(\Omega) \to L^2(\Omega)$ is C^2-mapping. For the Banach spaces $X = Y = H_0^1(\Omega)$ and $Y' = H^{-1}(\Omega)$, $F : X \to Y'$ is given by

$$\begin{aligned} &\text{for all } \ u, v \in H_0^1(\Omega) \\ &\langle F(u), v\rangle_{Y'Y} = \int_\Omega \nabla u \cdot \nabla v dx + \int_\Omega G(u) v dx. \end{aligned} \tag{51}$$

We assume that there exists a unique solution $u \in H_0^1(\Omega) \cap H^2(\Omega)$ to the problem

$$F(u) = 0$$

and that $DF(u) \in \mathscr{L}(X; Y')$ is an isomorphism. Furthermore, we assume that the bilinear form $b_u : X \times X \to \mathbb{R}$,

$$b_u(v, w) = \int_\Omega \nabla v \cdot \nabla w dx + \int_\Omega DG(u) v w dx$$

satisfies inf-sup condition (42).

Given $h \in (0, 1)$, we assume that $X_h = Y_h \subset X$ is a piecewise linear finite element space defined on a quasi-uniform triangulation of Ω satisfying

$$\inf_{w \in X_h} \{\|v - w\|_{0,p} + h\|v - w\|_{1,p}\} \le Ch^2 \|v\|_{2,p}, \forall v \in W_p^2(\Omega) \cap H_0^1(\Omega), 1 \le p \le \infty. \tag{52}$$

A Galerkin approximation of the problem (51) is to find $u^h \in X_h$ such that

$$\forall v^h \in X_h, \quad \left\langle F(u^h), v^h \right\rangle_{Y'Y} = \int_\Omega \nabla u^h \cdot \nabla v^h dx + \int_\Omega G(u^h) v^h dx = 0. \tag{53}$$

Since assumptions of Theorem 4.3 are satisfied, there exists a unique discrete solution u^h of (53). Furthermore, it can be proven that (cf. [12]) if h is sufficiently small, then u^h satisfies

$$\|u - u^h\|_{0,p} + h\|u - u^h\|_{1,p} \le Ch^2 \|u\|_{2,p}, \quad \forall 2 \le p \le \infty. \tag{54}$$

If D^2G is bounded on all bounded subsets of X and the exact solution u is in $W_p^2(\Omega)$ with $p > 2$ for $d = 2$and $p = \frac{12}{5}$ for $d = 3$, then we have the following improved result:

$$\|u_h - u^h\|_X \le CH^4,$$

where u_h is the solution obtained in **Step 2** of **Algorithm 3**. In fact, in the proof of Theorem 4.3, we have for all $v \in Y_h$,

$$\begin{aligned} b_{\phi^H}(u_h - u^h, v) &= \int_\Omega \Big[G(u^h) - G(u^H) - DG(u^H)(u^h - u^H)\Big] v dx \\ &= \int_\Omega M(u^h - u^H)^2 v dx, \end{aligned}$$

where $M = \int_0^1 (1-t) D^2 G(u^H + t(u^h - u^H))\, dt \in \mathscr{L}_2(X; Y')$. By assumption of boundedness of D^2G and (54), M is a uniformly (with respect to both H and h) bounded function on $\bar{\Omega}$. Using the Hölder inequality and the Sobolev inequalities $\|u\|_{0,p} \le C\|u\|_{1,2}$ ($1 \le p < \infty$ for $d = 2$ and $p = 6$ for $d = 3$), we deduce that

$$\int_\Omega (u^h - u^H)^2 v dx \le \|(u^h - u^H)^2\|_{0,\frac{p}{2}} \|v\|_{0,\frac{p}{p-2}} \le C\|u^h - u^H\|_{0,p}^2 \|v\|_{1,2}$$

It follows from Lemma 4.2 and (54) that

$$\|u_h - u^h\|_{1,2} \le C\|u^h - u^H\|_{0,p}^2 \le CH^4.$$

Moreover, we assume that G is C^3-mapping and D^3G is bounded on all bounded subsets of X and the projection $\Pi_H \in \mathcal{L}(Y; Y_H)$ defined in (47) satisfies for $k = 1, 2$

$$\|w - \Pi_H w\|_{0,2} + H\|w - \Pi_H w\|_{1,2} \le CH^k \|w\|_{k,2}, \quad \forall w \in H_0^1(\Omega) \cap H^k(\Omega).$$

By similar arguments, if $u \in W_4^2(\Omega)$ is the solution of (51), $u^h \in X_h$ is the solution of (53) and u_h^* is the solution obtained from **Algorithm 4** or **Algorithm 5**, then we can get quite improved estimates (cf. [22])

$$\|u_h^* - u^h\|_{1,2} \le CH^5, \qquad \|u_h^* - u^h\|_{0,2} \le CH^6.$$

From (54), we conclude that

$$\|u - u_h^*\|_{1,2} \le C(h + H^5), \qquad \|u - u_h^*\|_{0,2} \le C(h^2 + H^6).$$

6 Numerical experiments

In this section, we present numerical results to show the performance of our algorithms (ALG1 - ALG5) in various situations. In particular, rough solutions as well as smooth solutions are considered for the test. All simulations are conducted on Matlab.

For semilinear problems considered in Example 1 and Example 2, our ALG1 and ALG2 coincide with ALG3 and ALG4, respectively. However, analysis in the previous

section shows that ALG3 and ALG4 allow higher order relation between coarse and fine meshes.

6.1 Example 1 (smooth solution)

We consider a semilinear problem (15) (or (50)) defined on the domain $\Omega = (0, 1) \times (0, 1)$ with $G(\lambda, u) = u^3 - f$, and the variational formulation (16) (or (51)). Note that if $f \in L^p(\Omega)$ for $p \geq 4$, then it can be easily showed that all assumptions mentioned in Sect. 5 hold for all $H \leq H_0$ with some positive H_0 in the standard piecewise linear conforming finite element space. For the test, the function f is chosen from the exact solution

$$u(x, y) = xy(1 - x)(1 - y).$$

Tables 1, 2, 3, 4 and 5 display computed order of convergence for each algorithm (ALG1 - ALG5) under the given relation between the coarse and fine meshes, which support the theory developed in this paper.

Table 1 Convergence order (CO) of ALG1

$h = H^2$ $1/h(1/H)$	$\|u - u_h\|_{0,2}$	CO	$h = H^2$ $1/h(1/H)$	$\|u - u_h\|_{1,2}$	CO
16(4)	3.6536e−04	*	16(4)	1.5185e−02	*
64(8)	2.2935e−05	2.00	64(8)	3.8032e−03	1.00
256(16)	1.4338e−06	2.00	256(16)	9.5090e−04	1.00

Table 2 Convergence order (CO) of ALG2

$h = H^3$ $1/h(1/H)$	$\|u - u_h^*\|_{0,2}$	CO	$h = H^3$ $1/h(1/H)$	$\|u - u_h^*\|_{1,2}$	CO
64(4)	2.2938e−05	*	64(4)	3.8032e−03	*
216(6)	2.0142e−06	2.00	216(6)	1.1270e−03	1.00
512(8)	3.5848e−07	2.00	512(8)	4.7545e−04	1.00

Table 3 Convergence order (CO) of ALG3

$h = H^2$ $1/h(1/H)$	$\|u - u_h\|_{0,2}$	CO	$h = H^4$ $1/h(1/H)$	$\|u - u_h\|_{1,2}$	CO
16(4)	3.6536e−04	*	81(3)	3.0051e−03	*
64(8)	2.2935e−05	2.00	256(4)	9.5090e−04	1.00
256(16)	1.4338e−06	2.00	625(5)	3.8949e−04	1.00

Table 4 Convergence order (CO) of ALG4

$h = H^3$			$h = H^5$		
$1/h(1/H)$	$\|u - u_h^*\|_{0,2}$	CO	$1/h(1/H)$	$\|u - u_h^*\|_{1,2}$	CO
64(4)	2.2938e−05	*	243(3)	1.0018e−03	*
216(6)	2.0142e−06	2.00	1024(4)	2.3773e−04	1.00
512(8)	3.5848e−07	2.00	3125(5)	7.7899e−05	1.00

Table 5 Convergence order (CO) of ALG5

$h = H^3$			$h = H^5$		
$1/h(1/H)$	$\|u - u_h^*\|_{0,2}$	CO	$1/h(1/H)$	$\|u - u_h^*\|_{1,2}$	CO
64(4)	2.2938e−05	*	243(3)	1.0018e−03	*
216(6)	2.0142e−06	2.00	1024(4)	2.3773e−04	1.00
512(8)	3.5848e−07	2.00	3125(5)	7.7899e−05	1.00

6.2 Example 2 (boundary layer)

We consider the same semilinear problem as in Example 1, but the exact solution u is given by

$$u(x, y) = xy(1 - x)(1 - y)\exp(10x)\exp(10y)/53793,$$

which possesses boundary layers near the point (1, 1).

Table 6 compares the relative errors of solutions obtained from various algorithms with respect to H^1-norm. In the first column of Table 6, $u - u^H$ and $u - u^h$ represent the errors of solutions obtained directly by nonlinear solver on coarse grid and on fine grid, respectively. The expressions $u - u_h$, $u - (u_h + e_h^*)$, and $u - (u_h + e_H^*)$ display the errors of the solutions corresponding to ALG3, ALG4, and ALG5, respectively. Fixing the fine grid with meshsize $h = 1/1024$, we change coarse grid meshsizes ($H = 1/4,\ 1/8,\ 1/16,\ 1/32$) to see the effect of an initial guess for one Newton iteration on the fine grid. Notice that if $H = 1/4$ and $H = 1/32$, relations $h = H^5$ and $h = H^2$ hold, respectively. In the case of smooth solutions, for any H algorithms ALG3 - ALG5 provide the same accuracy as nonlinear solver on fine grid. However, for solutions with boundary layer, if H is too coarse (eg, $H = 1/4,\ 1/8$), the solutions of ALG3 - ALG5 are not as accurate as that of nonlinear solver. This indicates that coarse grid approximations to boundary layers should be good enough to be the initial guess for one Newton iteration on the fine grid. One the other hand, for $H = 1/16$, the correction step makes more sense. Note that correction on fine mesh provides more accurate solution than correction on coarse mesh.

Table 6 Various errors of H^1-norm for Examples 1 and 2

	Example 1 $1/H = 4$	$1/H = 8$	$1/H = 16$	$1/H = 32$	Example 2 $1/H = 4$	$1/H = 8$	$1/H = 16$	$1/H = 32$
$u - u^H$	4.27e−01	2.07e−01	1.02e−01	5.11e−02	1.26e+00	8.73e−01	4.98e−01	2.60e−01
$u - u_h$	1.59e−03	1.59e−03	1.59e−03	1.59e−03	3.01e−01	7.26e−02	1.06e−02	8.26e−03
$u - (u_h + e_h^*)$	1.59e−03	1.59e−03	1.59e−03	1.59e−03	7.46e−02	1.09e−02	8.26e−03	8.25e−03
$u - (u_h + e_H^*)$	1.59e−03	1.59e−03	1.59e−03	1.59e−03	2.81e−01	5.15e−02	9.11e−03	8.25e−03
$u - u^h$	1.59e−03	1.59e−03	1.59e−03	1.59e−03	8.25e−03	8.25e−03	8.25e−03	8.25e−03

6.3 Example 3 (singular solution)

In this example, we investigate the performance of ALG1 and ALG2 for nonlinear problems with insufficient regularity. We consider an elliptic problem (20) with gradient nonlinearities such that

$$\begin{aligned} -\mathrm{div}(\nabla p) + b(x, p, \nabla p) = 0 \quad &\text{in} \quad \Omega, \\ p(x) = -g \quad &\text{on} \quad \partial\Omega, \end{aligned}$$

where $b(x, p, \nabla p) = p^2 + |\nabla p|^2 - b_0(x)$ for a function $b_0(x)$ and Ω is the L-shape domain $\Omega = (-1, 1) \times (-1, 1) \setminus [0, 1] \times [0, -1]$. The functions b_0 and g are computed by the exact solution p of polar form

$$p(r, \theta) = r^{2/3} \sin(2\theta/3).$$

The lowest order Raviart–Thomas space is used on uniform triangular meshes for the mixed formulation (24). Note that the flux variable $\boldsymbol{u} = -\nabla p$ is singular at the origin. Let $m = 4$, that is, $X = L^4(\Omega) \times L^4(\Omega)^2$. For the relation $h = H^2$, Table 7 shows the errors and estimated orders of convergence for ALG1 and ALG2, Nonlinear Solver. Here, nonlinear solver is based on a simple fixed point iteration.

Since the norm of X is a product norm of L^4-spaces and the domain is an L-shape, for the nonlinear solution, we expect that

$$\|u - u^h\|_X \le Ch^{2/3-1/2-\epsilon} = Ch^{1/6-\epsilon},$$

for any $\epsilon > 0$. Indeed, since D^2G and D^3G are bounded on all bounded subsets of $\Lambda \times X$, we can derive that

$$\|u - u_h\|_X \le Ch^{1/6-\epsilon} \quad \text{and} \quad \|u - u_h^*\|_X \le Ch^{1/6-\epsilon},$$

which is illustrated numerically in Table 7.

7 Concluding remarks

We have developed the two-grid method proposed by Xu in the abstract framework of Brezzi, Rappaz, and Raviart (BRR) and in its particular framework with duality

Table 7 Convergence order for Example 3

$1/h(1/H)$	ALG1 $\|u - u_h\|_X$	CO	ALG2 $\|u - u_h^*\|_X$	CO	Nonlinear solver $\|u - u^h\|_X$	CO
16(4)	2.2166e−01	*	2.2046e−01	*	2.2054e−01	*
64(8)	1.7506e−01	0.170	1.7450e−01	0.169	1.7453e−01	0.169
256(16)	1.3856e−01	0.169	1.3839e−01	0.167	1.3840e−01	0.167

pairing proposed by Caloz and Rappaz. Our convergence results indicate that the relation between fine and coarse meshes is given by $h = \mathcal{O}(H^2)$ for all the nonlinear problems which can be written in and applied to either framework.

Note that the BRR framework with duality pairing allows higher order relations if it is applied to a particular example. For the semilinear problem, ALG3 allows the scaling $h = \mathcal{O}(H^4)$ with respect to H^1-norm, which follows from the smoothness conditions of nonlinear problems. Indeed, under the hypothesis (H1) (i.e., Lipschitz continuity of $D_\phi G$), the usual scaling $h = \mathcal{O}(H^2)$ is obtained. However, with the stronger assumptions as in Xu's work [22] (eg., G is a C^2-mapping and the exact solution has a higher regularity), the higher order relation $h = \mathcal{O}(H^4)$ can be derived. Also, the scaling $h = \mathcal{O}(H^5)$ can be achieved with an extra correction either on fine mesh (ALG4) or coarse mesh (ALG5).

Note also that the relation between coarse and fine meshes can be hard to hold for non-uniform or unstructured meshes, in which case, fine grids can be obtained by a few regular refinements from given coarse grids and two-grid algorithms can then be applied to produce comparable results; see [19] for more details on this idea. Finally, we mention that, by no means, all the existing two-grid methods can be applied to our framework. Our unified approach facilitates, within the framework, error analysis as well as unique solvability of two-grid methods for some class of nonlinear problems.

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